

SIMATS ENGINEERING ASSESSMENT TOOL

COURSE NAME: Theory of Computation COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO1: Analyze fundamental mathematical concepts of automata theory for formal languages and decision problems(BL:3)

Assessment Tool Weightage: 10%

QUESTIONS: 4 Total Marks:20 Duration : 1 Hour

Technical Presentation

Code of Conduct:

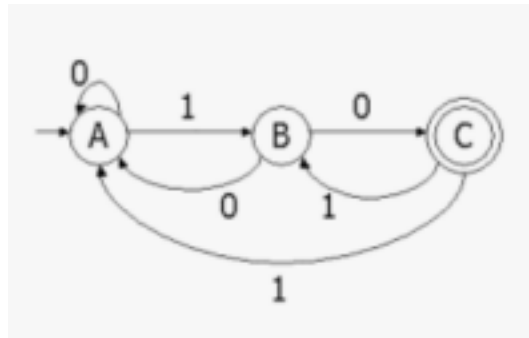
I _Pusalapati Chandu (192372119)___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Pchandu

1. How can every Non-Deterministic Finite Automaton (NFA) be converted into an equivalent Deterministic Finite Automaton (DFA), and what are the challenges involved? Consider the NFA given below which accepts some language L over the alphabet $\Sigma = \{0,1\}$. Construct a DFA that represents the same language L



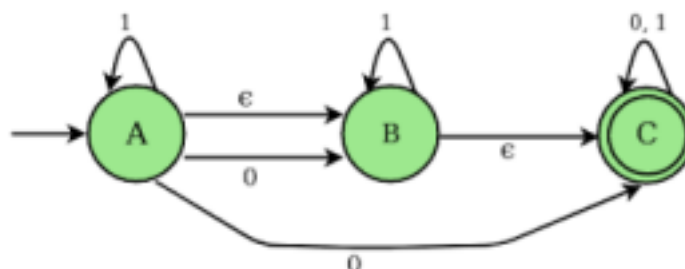
2. How is mathematical induction used to prove properties of strings and languages in automata theory? Apply the principle of mathematical induction and prove that for all positive integers n , the statement given below is true.

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

3. What role do epsilon (ϵ) transitions play in finite automata, and how can ϵ -NFA be converted into an equivalent NFA without epsilon transitions? Construct an equivalent NFA without ϵ -transitions that accepts the same language. Draw its transition table and mark the initial and final states



4. A NFA with ϵ -transitions is given below. Find ϵ -closure for each state and construct an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final states





SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105.

Smart ATM Transaction Validation System Using State Machines

Student Name – D Guna Vardhan
Register Number - 192372041

Student Name - A Dileep Kumar
Register Number- 192372134

Student Name - Sanjay Kumar Soy
Register Number- 192311151

Student Name - Pusalapati Chandu
Register Number- 192372119

Supervisor
Dr K Baskar, Professor
Department of Computer Science and Engineering
SIMATS Engineering

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Abstract

- ATMs are critical banking touchpoints, but weak validation of the transaction sequence can allow invalid or fraudulent operations to slip through.
- This project models valid ATM sessions using Finite State Machines (FSM), a core concept from Theory of Computation, so that only well-formed sequences of actions are ever accepted.
- Objective: build a validator that accepts card insertion, PIN entry, operation selection and confirmation only in the correct order, and rejects any illegal transition.
- Java implements the FSM engine and application logic; PostgreSQL stores accounts, transactions and audit logs.
- Expected outcome: a working prototype that validates ATM sessions in real time, blocks out-of-sequence or unauthorized actions, and logs every state change for auditing.

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Problem Statement

- Most ATM systems validate individual inputs, such as PIN or amount, but not the overall sequence of operations - for logically invalid or replayed transaction flows.
- Repeated invalid PIN attempts are often handled with ad-hoc counters instead of a formal lock-out state, weakening security.
- Hand-written if-else validation logic is difficult to test, extend, and prove correct as more transaction types are added.
- A formal, FSM-based validator is needed so that every transaction path can be verified against a clearly defined set of states and allowed transitions.

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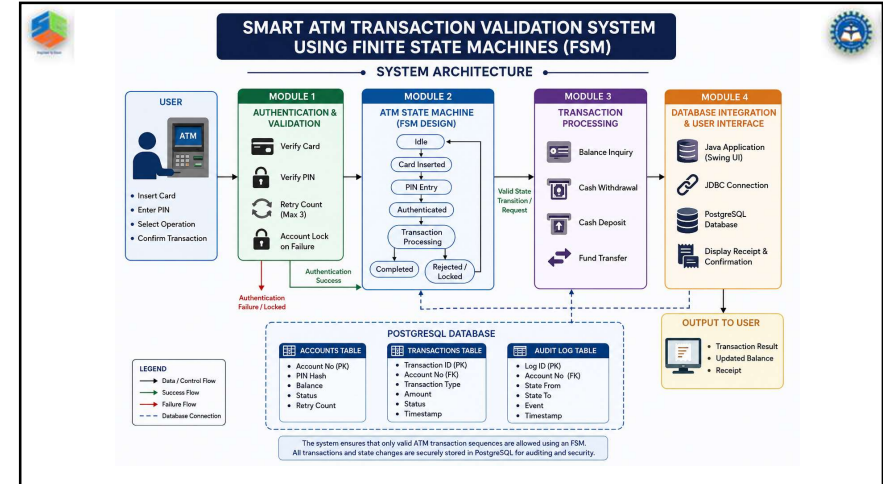
Objectives

- To design and formally specify a Finite State Machine that models every valid ATM transaction state and transition - Idle, Authentication, Transaction Selection, Processing, and Completion/Rejection.
- To implement this FSM in Java, integrated with a PostgreSQL backend, so that only valid transaction sequences are authenticated, processed, and logged.

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Literature Survey					
SL No	Year of Publication	Authors	Title of the Publication	Findings	Limitations
1	2023	Kumar et al.	Finite State Machine Based Approach for Secure ATM Transaction Validation	FSM model reduced invalid/out-of-sequence transactions in simulation	Tested only in a single-bank scenario; no multi-user concurrency
2	2022	Sharma & Ran	Formal Verification of Banking Protocols Using State Machines	Model checking detected several insecure state transitions	High computation cost for larger state spaces
3	2021	Ahmed et al.	A Survey on ATM Security Threats and Countermeasures	Categorized common ATM threat vectors and existing defenses	Survey only; no implemented or tested solution
4	2023	Fernandes & Silva	Java-Based Transaction Processing with a Relational Database Backend	Demonstrated reliable ACID transaction handling for banking apps	Did not address sequence-level (state) validation
5	2024	Nair et al.	Authentication State Modeling for Financial Self-Service Terminals	Modeled PIN retry and lock-out behaviour as finite states	Not evaluated against real ATM hardware

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Module 1: User Authentication & Validation	
● Purpose:	verify the ATM card and PIN before any transaction is permitted, applying retry-limit and lock-out rules.
● Inputs:	card number, entered PIN, and the stored credential hash from the accounts table.
● Process:	read card -> prompt PIN -> compare against the stored hash -> allow on match, increment the retry counter on mismatch, lock the account after 3 failed attempts.
● Output:	a validated (or denied/locked) user session, which is handed to Module 2 for state-sequence control.
● Role in the system:	the first checkpoint - no request reaches the State Machine unless the user's identity is confirmed here.

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Module 2: ATM State Machine Design – Overview	
● Purpose:	model every valid ATM session as a Finite State Machine (FSM) so that only correctly ordered actions are ever accepted.
● States:	Idle, Card Inserted, PIN Entry, Authenticated, Transaction Processing, Completed, and Locked/Rejected.
● Inputs:	the authentication result from Module 1 and transaction requests from Module 3.
● Output:	the current valid state and an accept/reject decision for every incoming action.
● Role in the system:	the central control module - every other module must consult it before performing any state-changing action.

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Module 2: ATM State Machine Design – Technical Details



- The State Transition Table (below) defines every allowed move; any input outside this table is rejected by the engine.
- Implemented in Java as a table-driven transition function keyed on (currentState, event) -> nextState.

Current State	Input	Next State
Idle	Card Inserted	Card Inserted
Card Inserted	PIN Entered (correct)	Authenticated
Card Inserted	PIN Entered (wrong, retries)	Card Inserted
Card Inserted	PIN Entered (wrong, retries)	Locked
Authenticated	Select Transaction	Transaction Processing
Transaction Processing	Operation Success/Failure	Completed / Declined

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Module 3: ATM Transaction Processing



- Purpose: carry out Balance Inquiry, Withdrawal, Deposit and Fund Transfer once the State Machine confirms an Authenticated/Transaction-ready state.
- Validation Algorithm: confirm state = Authenticated -> validate requested amount/account -> update balance -> transition to Completed or Declined.
- Every operation is written to the transactions table with a timestamp, type, amount and resulting status.
- Ensures ACID consistency so no partial or duplicate transaction is ever recorded, even if an error occurs mid-operation.

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Module 4: Database Integration & User Interface



- Database: PostgreSQL stores three core tables - accounts (account_no, balance, pin_hash), transactions (txn_id, account_no, type, amount, status, timestamp), and audit_log (state transitions).
- Integration: the Java application connects to PostgreSQL via JDBC; every state change and transaction is persisted for auditing.
- User Interface: a simple ATM screen flow - Card/PIN entry -> Menu (Balance / Withdraw / Deposit / Transfer) -> Confirmation & Receipt.
- Role in the system: provides the persistence layer and the screens the user interacts with, tying Modules 1-3 together into one working application.

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Implementation & Output



- Module 1 output: Login and PIN verification screens validate the user before any menu is shown.
- Module 2 output: console/log trace showing the FSM's state transitions for a sample session (accepted and rejected paths).
- Module 3 output: transaction menu and confirmation screens that update the balance instantly and generate a receipt.
- Module 4 output: PostgreSQL tables showing the persisted accounts, transactions and audit trail after a completed session.
- Insert your actual Login, Menu, Transaction, Receipt, and Database Table screenshots on this slide.

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Testing & Evaluation

- Functional Testing: verified each operation (balance inquiry, withdrawal, deposit, transfer) returns the expected result.
- State Transition Testing: confirmed the FSM (Module 2) rejects every out-of-sequence input, e.g. a withdrawal attempted before authentication.
- Authentication Testing: validated PIN retry limits and the account lock-out behaviour after repeated failures (Module 1).
- Database Testing: checked transaction consistency, correct balance updates, and rollback on a failed operation (Module 4).
- Performance Metrics: average response time, validation accuracy, and false-accept / false-reject rate were recorded across the test transactions.

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SDG Contribution

SDG 9 – Industry, Innovation and Infrastructure

- Develops a secure ATM transaction validation system using Finite State Machines.
- Enhances digital banking infrastructure through reliable transaction processing.
- Promotes innovation in software-based financial systems.
- Improves the efficiency and reliability of ATM operations.

SDG 16 – Peace, Justice and Strong Institutions

- Ensures secure user authentication and transaction validation.
- Reduces fraud through controlled state transitions.
- Enhances trust in digital banking services.
- Supports secure and accountable financial institutions.



Conclusion & Future Work

- Conclusion: the four modules together - Authentication & Validation, State Machine Design, Transaction Processing, and Database Integration & UI - enforce correct transaction sequencing, improve authentication reliability, and keep an auditable trail of every ATM session.
- Future Scope: biometric authentication and facial recognition for identity verification, AI-based fraud/anomaly detection, mobile banking integration, and migration to a cloud-hosted database.

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References

- [1] J. E. Hopcroft, R. Motwani, and J. D. Ullman, Introduction to Automata Theory, Languages, and Computation, 3rd ed., Pearson, 2006.
- [2] Kumar et al., "Finite State Machine Based Approach for Secure ATM Transaction Validation," sample entry - replace with your verified source, 2023.
- [3] PostgreSQL Global Development Group, PostgreSQL 16 Documentation. [Online]. Available: <https://www.postgresql.org/docs/>
- [4] Oracle Corporation, Java Platform, Standard Edition Documentation. [Online]. Available: <https://docs.oracle.com/en/java/>



Thank You

